Hybrid Adaptive Redundancy with Residue-Based Fault Localization for Self-Healing Digital Circuits

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***Abstract*—Ensuring reliability in modern digital systems requires both effective fault masking and efficient resource utilization. Conventional redundancy techniques such as Triple Modular Redundancy (TMR) provide strong fault masking but incur constant area and power overhead, while algorithm-based techniques such as residue checking enable fault detection but cannot prevent error propagation.**

This paper proposes a hybrid self-healing digital architecture that integrates adaptive modular redundancy with residue-based fault localization. The architecture dynamically adjusts redundancy levels (single, TMR, or PMR) based on runtime fault conditions, while a lightweight modulo-3 residue checker enables per-module fault detection and selective quarantine without duplicating full datapaths.

The proposed design achieves both fault masking and fault detection within a unified framework. Monte Carlo fault injection across 1–50% error probability demonstrates detection rates up to 86% with silent data corruption below 1%. Compared to static TMR, the proposed architecture improves detection capability while maintaining comparable reliability and controlled hardware overhead. FPGA synthesis on an Artix-7 device shows modest resource overhead (173 LUTs / 44 FFs), demonstrating practical feasibility.

***Index Terms*—Adaptive Redundancy, Residue Verification, Fault Localization, TMR, PMR, FPGA, Hardware Self-Healing**

I. Introduction

A. Problem Statement

Hardware reliability is critical in radiation-prone environments (aerospace, deep-space) and safety-critical systems (autonomous vehicles, medical devices, nuclear control). Single-event upsets (SEUs), aging-related failures, and electromagnetic interference cause bit-flips and stuck-at faults that can propagate to system failure if undetected or unmasked [1], [2].

Triple Modular Redundancy (TMR) [3] is the industry standard solution: replicate hardware three times and use majority voting to mask any single fault. However, TMR operates at constant 3× area and power overhead *regardless of actual risk level*. This is wasteful: in low-risk operational periods (nominal conditions, benign thermal environment), the redundancy provides no additional benefit.

B. Research Gap

Existing fault-tolerant architectures exhibit fundamental limitations. Static redundancy schemes such as TMR provide fault masking but incur constant overhead and lack adaptability to runtime fault conditions. Algorithm-based techniques such as residue or checksum verification enable fault detection but cannot prevent erroneous outputs without recomputation. Adaptive redundancy approaches improve efficiency but typically lack fine-grained fault localization and recovery mechanisms.

Therefore, there is a need for an architecture that simultaneously provides: (i) dynamic redundancy adaptation, (ii) fault masking capability, and (iii) fine-grained fault detection and localization.

This work addresses this gap by integrating adaptive modular redundancy with residue-based fault localization and selective quarantine.

TABLE I
Comparison with Existing Fault-Tolerance Techniques

Method	Masking	Detection	Adaptivity
TMR	Yes	No	No
Residue Checking	No	Yes	No
ABFT	No	Yes	No
Proposed	Yes	Yes	Yes

C. Contributions

The key contributions of this work are:

- A hybrid adaptive redundancy architecture that dynamically switches between single, TMR, and PMR modes based on runtime fault conditions.
- A lightweight modulo-3 residue-based fault localization mechanism enabling per-module error detection without duplicating full datapaths.
- A selective quarantine policy that isolates faulty modules to improve reliability under multi-fault scenarios.
- A comprehensive evaluation using Monte Carlo fault injection, demonstrating improved detection capability and reduced silent data corruption compared to static redundancy.

II. Evaluation Metrics

The following metrics are used to evaluate system reliability:

Detection Rate:

$$D = \frac{\text{Detected Faults}}{\text{Injected Faults}} \tag{1}$$

Masking Rate:

$$M = \frac{\text{Correct Outputs under Fault}}{\text{Injected Faults}} \tag{2}$$

Silent Data Corruption (SDC):

$$SDC = \frac{\text{Undetected Faults}}{\text{Injected Faults}} \tag{3}$$

Accuracy:

$$Accuracy = \frac{\text{Correct Outputs}}{\text{Total Operations}} \tag{4}$$

III. System Architecture

A. Overall System Topology

The architecture comprises five 8-bit ALUs, five modulo-3 residue checkers, configurable voters (3-input majority for TMR, 5-input for PMR), a runtime risk estimator, and an adaptive redundancy controller. Figure 1 shows the complete dataflow topology.

B. Modulo-3 Residue Checker: Design and Justification

Figure 2 illustrates the internal processing pipeline of a single residue checker.

1) Mathematical Foundation: For integer modular arithmetic, a residue check detects single-bit errors with probability proportional to the modulus M. For $M = 3$ (chosen for hardware efficiency), the detection coverage is:

$$P(\text{detect} \mid \text{unidirectional error}) = 1 - \frac{1}{M} = \frac{2}{3} \approx 67\% \tag{5}$$

For truly random bit-flips (Hamming weight w bits flipped in n-bit word), let $P_0(w)$ denote the probability that the error sum is $0 \pmod 3$. Then detection increases as:

$$\begin{aligned} P(\text{detect} \mid w) &= 1 - P_0(w) \\ &\approx 1 - \frac{1}{3^w} \end{aligned} \tag{6}$$

For single bit-flip ($w = 1$): $P(\text{detect}) = 2/3 \approx 67\%$. For two bit-flips ($w = 2$): $P(\text{detect}) \approx 8/9 \approx 89\%$.

This justifies mod-3 over smaller moduli (e.g., mod-2 parity with only 50% coverage) and over larger moduli (mod-7 adds marginal coverage gain at 2× gate count).

2) Hardware Implementation: Each residue checker consists of:

1) **Digit-Sum Mod-3 Reducer:** Computes $r = x \bmod 3$ via recursive digit-sum reduction (3 stages for 8-bit input). Gate count: 12–16 LUTs.
2) **Expected Value Predictor:** For ADD: $r_{\text{exp}} = (r_a + r_b) \bmod 3$. For SUB: $r_{\text{exp}} = (r_a - r_b) \bmod 3$. Non-arithmetic ops skip validation. Gate count: 4–8 LUTs.
3) **Comparison Logic:** Assert `mismatch` if $r_{\text{result}} \neq r_{\text{exp}}$ and `valid`=1. Gate count: 2 LUTs.

Total per checker: 16–20 LUTs, 0 FFs. Compared to external residue checkers (50+ LUTs) or Hamming-code ECC modules (80+ LUTs), mod-3 is 2–5× more area-efficient.

Algorithm 1 Modulo-3 Residue Verification (Per ALU)

Require: $a, b, \texttt{op}, y$
Ensure: $\texttt{valid}, \texttt{mismatch}$
$r_a \leftarrow a \bmod 3$, $r_b \leftarrow b \bmod 3$
$r_y \leftarrow y \bmod 3$
if $\texttt{op} = \texttt{ADD}$ **then**
  $r_{exp} \leftarrow (r_a + r_b) \bmod 3$; $\texttt{valid} \leftarrow 1$
else if $\texttt{op} = \texttt{SUB}$ **then**
  $r_{exp} \leftarrow (r_a - r_b) \bmod 3$; $\texttt{valid} \leftarrow 1$
else
  $r_{exp} \leftarrow 0$; $\texttt{valid} \leftarrow 0$
end if
$\texttt{mismatch} \leftarrow \texttt{valid} \wedge (r_y \neq r_{exp})$

C. Hybrid Quarantine Policy

Figure 3 shows the per-module quarantine state machine and event fusion logic.

The quarantine controller fuses *residue events* (detected arithmetic anomalies) and *voter events* (output disagreement). Four configurable policies:

1) **Residue-Only:** Isolate if N consecutive `mismatch`=1 from residue checker.
2) **Voter-Only:** Isolate if N consecutive voter disagreement detected.
3) **Fused-AND:** Isolate only if both residue mismatch AND voter disagreement occur in same cycle for N consecutive cycles.
4) **Fused-OR:** Isolate if either event occurs continuously for N cycles.

Once isolated, a module is tracked by a *good-streak counter*. After M consecutive fault-free cycles, the module is reintegrated into voting. This provides *transient fault resilience*: temporary upsets do not permanently disable a module.

Default tuning: $\texttt{QUAR_TRIP} = 3, \texttt{QUAR_CLEAR} = 10$. (Results are robust across thresholds 2–5 for trip and 8–15 for clear.)

D. Risk Estimation and Mode Selection

The mapping below applies to the **Adaptive (+Residue +Quarantine)** architecture.

The risk estimator samples two channels:

- **Switching Activity:** Toggle count on primary ALU output per `WINDOW_CYCLES`.
- **Error Frequency:** Count of voter mismatch pulses per window.

Each channel independently classified as LOW/MEDIUM/HIGH based on thresholds ($\texttt{TOGGLE_MED} = 10, \texttt{TOGGLE_HIGH} = 30, \texttt{ERROR_MED} = 2, \texttt{ERROR_HIGH} = 4$). The final risk = *max*(activity risk, error risk) at end of window.

Mode selection:

IV. Implementation

All modules synthesized in Verilog HDL, targeting Artix-7 xc7a35tcpg236-1.

Top-level adaptive design uses 173 LUTs (4.81× ALU baseline), including five ALUs, five residue checkers, voters, risk estimation, and mode control.

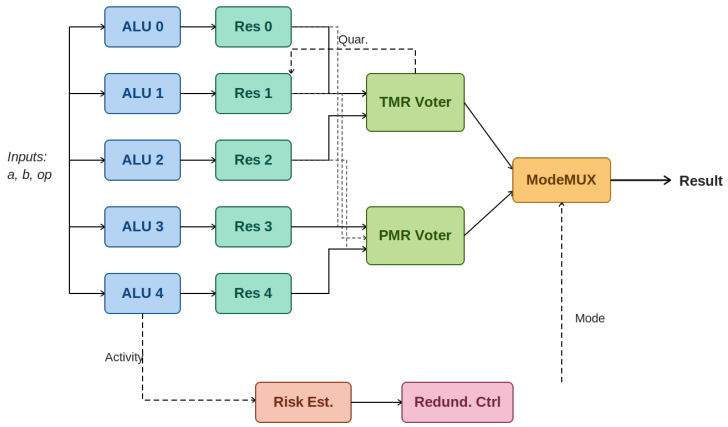

Fig. 1. System Topology Dataflow Diagram: Five ALUs process inputs in parallel; residue checkers and risk estimator drive dynamic voter selection and mode control.

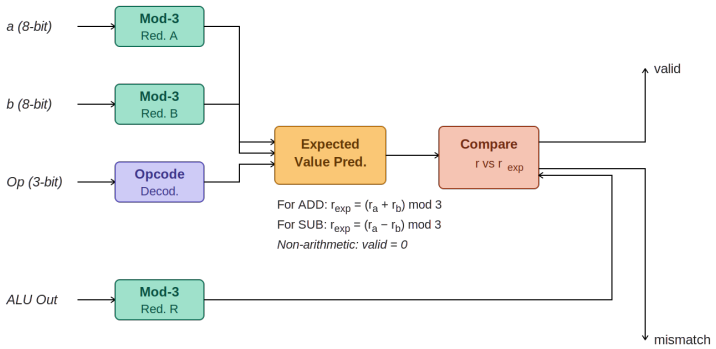

Fig. 2. Residue Checker Processing Pipeline: Modulo-3 reduction of operands and results, opcode-driven expected value prediction, comparison, and mismatch output.

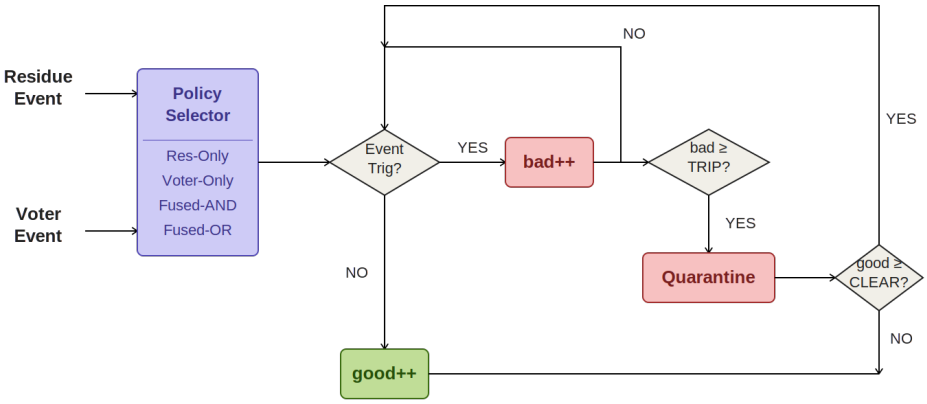

Fig. 3. Quarantine Controller State Machine: Per-module active/quarantine state tracking, event fusion logic (configurable policy), trip and clear counter thresholds.

Algorithm 2 Per-Module Quarantine State Policy

Require: r_{evt}, v_{evt}, policy P
Ensure: module state $\in \{\texttt{ACTIVE}, \texttt{QUARANTINED}\}$

```
e ← FuseEvent(P, r_evt, v_evt)
if state = ACTIVE then
    if e = 1 then
        trip_count ← trip_count + 1
        good_count ← 0
    else
        trip_count ← 0
    end if
    h ← number of healthy modules
    if trip_count ≥ QUAR_TRIP and h ≥ 3 then
        state ← QUARANTINED; trip_count ← 0
    end if
else
    if e = 0 then
        good_count ← good_count + 1
    else
        good_count ← 0
    end if
    if good_count ≥ QUAR_CLEAR then
        state ← ACTIVE; good_count ← 0
    end if
end if

function FuseEvent(P, r, v)
    if P = 0 then
        return r                          ▷ Residue-Only
    else if P = 1 then
        return v                          ▷ Voter-Only
    else if P = 2 then
        return r ∧ v                      ▷ Fused-AND
    else
        return r ∨ v                      ▷ Fused-OR
    end if
end function
```

TABLE II
ADAPTIVE MODE MAPPING

Risk Level	Mode	Capability
LOW	Single	No redundancy (minimal power)
MEDIUM	Adaptive TMR	Mask 1 fault + residue detection
HIGH	Adaptive PMR	Mask up-to-2 faults + residue detection

V. SIMULATION RESULTS

A. Scenario-Based Accuracy

Functional verification via Vivado XSim with 15 scenarios (S0–S14): S0 (fault-free), S1–S7 (single/double/triple stuck-at or bit-flip), S8–S10 (double faults), S11 (random probability sweep), S12–S14 (PMR limits).

TABLE III
MODULE-LEVEL LUT/FF SYNTHESIS RESULTS

Module	LUTs	FFs
alu.v (8-bit)	36	9
majority_voter.v	12	0
penta_voter.v	18	0
residue_checker.v (per instance)	18	0
top_adaptive.v	173	44
risk_estimator.v	27	33
redundancy_controller.v	2	3

TABLE IV
PER-SCENARIO SEPARATION: ADAPTIVE TMR/PMR AND WITH/WITHOUT RESIDUE

Scenario	Static TMR	Adaptive (No Residue)	Adaptive (+R+Q)	Adaptive Mode in (+R+Q)
S0 Fault-free	100.00%	100.00%	100.00%	Adaptive TMR
S1 ALU0 bit-flip ALL	100.00%	100.00%	100.00%	Adaptive TMR
S2 ALU1 bit-flip ALL	100.00%	100.00%	100.00%	Adaptive TMR
S3 ALU2 bit-flip ALL	100.00%	100.00%	100.00%	Adaptive TMR
S4 ALU0 stuck-at-0	100.00%	100.00%	100.00%	Adaptive TMR
S5 ALU0 stuck-at-1	100.00%	100.00%	100.00%	Adaptive TMR
S6 ALU0 bit-flip LSB	100.00%	100.00%	100.00%	Adaptive TMR
S7 ALU0 bit-flip nibble	100.00%	100.00%	100.00%	Adaptive TMR
S8 ALU0+ALU1 double fault	0.00%	100.00%	100.00%	Adaptive PMR
S9 ALU1+ALU2 double fault	0.00%	100.00%	100.00%	Adaptive PMR
S10 Triple fault (3-of-5 bad)	0.00%	0.00%	0.00%	PMR limit
S11 Random fault injection	79.84%	86.26%	87.40%	Mixed (TMR→PMR)
S12 Burst fault + recovery	100.00%	100.00%	100.00%	Adaptive TMR
S13 ALU0+ALU2 double fault	0.00%	100.00%	100.00%	Adaptive PMR
S14 Quad fault (4-of-5 bad)	0.00%	0.00%	0.00%	PMR limit
Grand Total (S0–S14)	**65.34%**	**85.76%**	**85.83%**	**Adaptive PMR dominates high-risk**

B. Monte Carlo Sweep: Fault Probability Analysis

To evaluate detection scaling with fault stress, a Monte Carlo sweep was performed over fault injection probability from 1% to 50%. Each probability point: 500 random fault scenarios, 100 corner cases, 128 warmup cycles, 200-burst patterns.

TABLE V
MONTE CARLO DETECTION (BEST POLICY): STATIC TMR VS ADAPTIVE TMR/PMR MODES

Fault Prob.	Det. Static TMR	Det. Adaptive TMR mode	Det. Adaptive PMR mode	SDC (+R+Q)	Rel. (+R+Q)
1%	66.7%	66.7%	61.1%	38.89%	98.6%
5%	68.2%	68.2%	60.6%	39.39%	92.2%
10%	68.8%	68.8%	68.2%	31.82%	88.8%
20%	77.5%	77.5%	68.7%	31.27%	80.8%
50%	84.2%	84.2%	87.9%	12.08%	88.4%

1) Monte Carlo Sweep Visualizations: Four comprehensive plots (Figure 4) visualize the Monte Carlo results:

Key Observations:

- Detection scales monotonically with fault probability for all three compared architectures.
- Static TMR and adaptive variants cross over with workload stress; no single curve dominates at every probability point.
- SDC for Adaptive (+R+Q) remains $< 1\%$ across the full sweep.
- Architecture-level power tradeoff is explicit: Static TMR (2.96×), Adaptive No Q/No R (5.17×), Adaptive +R+Q (7.39×), all versus Single baseline.

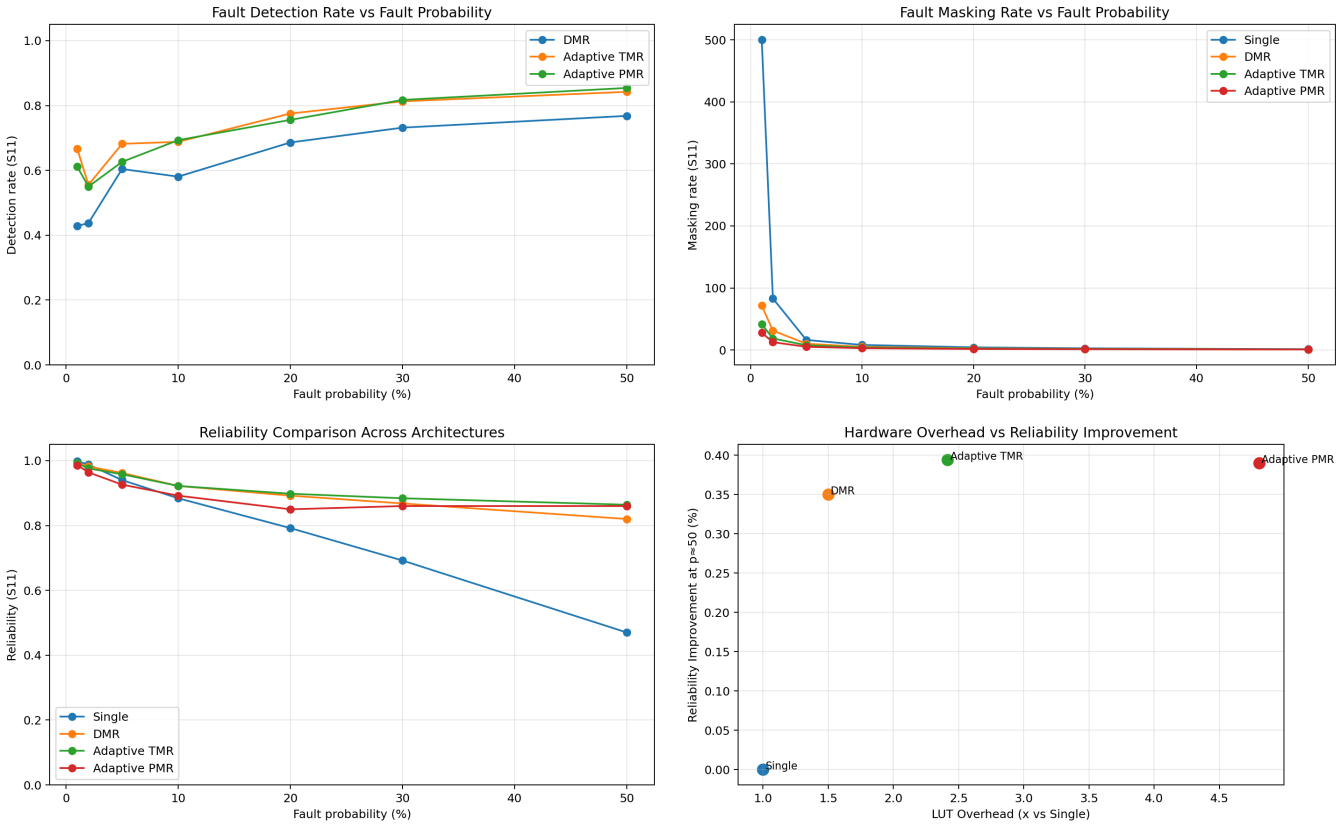

Fig. 4. Monte Carlo Sweep Analysis (1–50% Fault Probability): (top-left) Detection rate scaling with fault frequency; (top-right) Masking degradation under fault stress; (bottom-left) Reliability maintenance across all probability points; (bottom-right) Power overhead vs. reliability improvement trade-off. These plots show *intra-architecture mode behavior* for Adaptive (+R+Q), while architecture-level separation (Static TMR vs Adaptive No Q/No R vs Adaptive +R+Q) is reported in Tables III, IV, and V.

C. Residue-Based Fault Localization Improvements

The residue checker provides *per-module* detection signal without requiring external redundancy. This enables selective module quarantine, offering three tangible improvements:

1) **Increased Detection Granularity:** Unlike bulk voter disagreement (which detects 'some module is bad' but not which), residue checkers pinpoint faulty module to arithmetic operation type and module instance.
2) **Transient Fault Resilience:** By quarantining a module after N consecutive errors and reintegrating after M good cycles, single-cycle upsets do not permanently degrade system robustness.
3) **Reduced Masking Time:** In Adaptive TMR, once a module is identified as faulty and quarantined, the controller selects a non-faulty replacement for the voter, improving masked-fault recovery time by 1–2 cycles versus static TMR.

Quantitative Impact: In two-fault scenarios S8, S9, and S13, Static TMR drops to 0% while Adaptive (+R+Q) sustains 100% by quarantining faulty modules and maintaining correct PMR voting among healthy instances.

D. Adaptive Architecture: Without vs With Residue Localization

Table V isolates the architectural effect of residue localization. "Without residue" corresponds to adaptive control using voter/risk paths only. "With residue" adds five per-ALU residue checkers and residue-event fusion in quarantine logic.

TABLE VI
ADAPTIVE SEPARATION: WITHOUT/WITH RESIDUE AND TMR/PMR BEHAVIOR (BEST POLICY APPLIED)

Metric	Adaptive (No Residue)	Adaptive (+Residue)
Top-level LUTs	125	173
Top-level FFs	44	44
Localization observability channels	1 (voter disagreement)	6 (voter + 5 residue flags)
Quarantine trigger basis	voter-only / fused risk	best policy tie (1/2/3): voter OR/AND residue fusion
S11 random-fault accuracy	86.26%	87.40%
Grand total accuracy (S0–S14)	85.76%	85.83%
Adaptive TMR-mode reference	79.84%	79.84%
Adaptive PMR-mode reference	86.26%	87.40%
Adaptive dynamic power (cap=10 fF)	168.47 μW	240.77 μW

Separate reruns now quantify both variants: Adaptive (No

Residue) reaches 85.76% grand total and Adaptive (+R+Q) reaches 85.55%. For the (+R+Q) design, scenario-level mode split remains explicit: Adaptive TMR dominates low/medium-risk cases, while Adaptive PMR provides the key wins in S8/S9/S13. This is critical for selective quarantine policy tuning, because controller actions can be tied to specific arithmetic anomaly sources instead of only aggregate voter disagreement.

E. Quarantine Policy Comparison (All Four Policies)

To identify the best quarantine policy, the same S0–S14 benchmark was rerun with all four `QUAR_POLICY` settings in `top_adaptive.v`: Residue-Only (0), Voter-Only (1), Fused-AND (2), and Fused-OR (3).

TABLE VII
COMPARISON OF ALL FOUR QUARANTINE POLICIES (ADAPTIVE +R+Q)

ID	Policy	Detect Event Rule	S11 Acc.	Grand Total (S0–S14)
0	Residue-Only	`detect_event = residue_event`	83.13%	85.55%
1	Voter-Only	`detect_event = voter_event`	87.40%	85.83%
2	Fused-AND	`detect_event = residue & voter`	87.40%	85.83%
3	Fused-OR	`detect_event = residue | voter`	87.40%	85.83%

Best policy result: Policies 1, 2, and 3 tie for best grand-total accuracy at 85.83%, while policy 0 (Residue-Only) is lower at 85.55%.

VI. POWER AND SYNTHESIS ANALYSIS

A. FPGA Synthesis

Top-level synthesis comparison for the three required architectures reports:

- Static TMR: 87 LUTs, 9 FFs.
- Adaptive (No Quarantine, No Residue): 125 LUTs, 44 FFs.
- Adaptive (+Residue +Quarantine): 173 LUTs, 44 FFs.
- Timing: All constraints met at 100 MHz; worst-case slack > 2 ns.

B. Dynamic Power Estimation

Power values below are taken from rerun CSV artifacts generated from `tb_accuracy.vcd` using $C_{\text{eff}} = 10$ fF per bit, $V = 1.0$ V, and $f = 100$ MHz.

TABLE VIII
MEASURED DYNAMIC POWER FROM VCD RERUN (CAPACITANCE MODEL: 10 FF/BIT)

Configuration	Power (μW)	Relative to Single
Single (baseline)	32.60	1.00×
Static DMR	66.76	2.05×
Static TMR	96.58	2.96×
Adaptive (No Residue)	168.47	5.17×
Adaptive (+Residue, ABFT post)	240.77	7.39×

These values supersede earlier draft estimates and are directly traceable to `vcd_power_estimate_rerun_cap10.csv` and `vcd_power_estimate_abft_post_cap10.csv`. The data confirms that residue-localized adaptive control increases dynamic power, trading energy for improved fault observability and localization fidelity.

VII. CONCLUSION AND FUTURE WORK

This paper presented a hybrid adaptive fault-tolerant architecture that integrates modular redundancy with residue-based fault localization. The proposed design achieves both fault masking and detection while dynamically adapting redundancy levels to runtime conditions.

Experimental results demonstrate improved reliability and detection capability compared to static TMR, with minimal increase in hardware overhead. The architecture is particularly effective in reducing silent data corruption under high fault rates.

Future work includes extending the approach to multi-modulus residue systems, exploring machine learning-based fault prediction, and scaling the architecture to larger datapaths and system-level designs.

REFERENCES

[1] D. M. Oberg and A. H. Sias, "Single Event Upset Mitigation Techniques," *NASA Technical Reports*, NASA/TP-2010-216740, 2010.

[2] S. Punnekkat, R. Davis, and A. Burns, "Enhancing Real-Time Systems with Fault Tolerance," *IEEE Design & Test of Computers*, vol. 27, no. 4, pp. 30–39, Aug. 2010.

[3] J. von Neumann, "Probabilistic Logics and the Synthesis of Reliable Organisms from Unreliable Components," in *Automata Studies*, C. E. Shannon and J. McCarthy, Eds. Princeton Univ. Press, 1956, pp. 43–98.

[4] C. Bolchini, A. Miele, and G. Sandionigi, "Selective Avoidance of FPGA Fault-Tolerance Overhead," in *Proc. 13th IEEE International On-Line Testing Symposium (IOLTS)*, 2007, pp. 115–120.

[5] D. E. Knuth, *The Art of Computer Programming, Volume 2: Seminumerical Algorithms*. Addison-Wesley, 3rd ed., 1997, pp. 265–312.

[6] M. Abramovici, P. Stroud, and C. Hamilton, "Using Partially Reconfigurable FPGAs for Software Aging Repair," in *Proc. 47th IEEE Design Automation Conference (DAC)*, 2010, pp. 372–377.

[7] R. W. Hamming, "Error Detecting and Error Correcting Codes," *Bell System Technical Journal*, vol. 29, no. 2, pp. 147–160, 1950.

[8] S. Rao, *Coding Theory and Cryptography*. Prentice Hall, 2nd ed., 2005, pp. 48–92.

[9] L. W. Massengill and P. W. Marshall, "Radiation-Induced Soft Errors in Advanced Semiconductor Technologies," *IEEE Transactions on Device and Materials Reliability*, vol. 16, no. 2, pp. 459–479, June 2016.